

Phase 0/1 Assertions (regression checklist)

PREPARED BY Energy Hippo implementation team

The known-value checks to run **once the Phase 0/1 specs are loaded** into EEMSuite. These make Northlake a regression asset, not just a demo. Each maps to an `expected_assertion` ([implementation/synthetic-source-model.pdf](#)). Exact numbers that depend on generated bill/interval values are expressed as **relationships** (reconciliations, counts, comparisons) that hold regardless of the generator's seed; fixed counts are stated outright.

Sources: [bill-charge-line-spec](#), [phase1-substrate-extras](#), [data/eem/northlake-company-calendar-v1.yaml](#), [data/eem/northlake-eem-permissions-v1.yaml](#), [data/security/northlake-eem-security-v1.yaml](#).

Phase 0 — foundation

id	assertion	expected
substrate_loads	A blank EEMSuite seed loads the substrate without error	hierarchy + users + units + calendar present
calendar_count	CompanyCalendar rows per company (CalendarTypeID 1)	24 each (2024-01...2025-12)
calendar_companies	companies with a calendar	northlake_university, cedar_row_apartments, northlake_thermal_plant
fiscal_gap_absent	CompanyCalendar for 2026-01	no row (import-rejection fixture)
import_rejected_no_period	a bill-import dated 2026-01	rejected: "Fiscal Year/Period NOT found!"
substrate_reports_renderer	rp_BuildingList, rp_PointsList, rp_IndexList, rp_HierarchyDetailsEditor	non-empty
approver_vs_viewer	bill QC approval	AccessLevel-3 group can approve; bill_entry (2) and report_viewer (1) cannot
node_scope_intra_company	nlu_science_center_scope group visibility	Science Center nodes only
cross_company_isolation	Cedar Row groups	see only Cedar Row site/buildings
disabled_user_no_auth	casey.holt	in membership, cannot authenticate
readonly_no_edit	report_viewer / audit_readonly / support_readonly	no AuthLevel ≥2 on editable objects
security_counts	groups / users / active	27 / 19 / 18

Phase 1 — consumption & cost

id	assertion	expected
bill_history_complete	consecutive monthly bills per active account	≥24 (2024-01...2025-12), minus deliberate fixtures
charge_lines_sum_to_total	per bill, non-informational lines	= total_cost (±\$0.01)
usage_lines_reconcile	per commodity, consumption-line qty	= bill usage quantity
tou_lines_sum_to_kwh	VED TOU lines	sum to total kWh; winter bills carry no peak line
tier_blocks_partition_usage	RCU water tiers	partition total kgal at 50/200 bounds
credit_is_negative	HOS net_metering_credit / rec_transfer	< 0 / = 0 (informational)
tax_on_subtotal	utility_users_tax	= tax_rate × sum(preceding lines)
rollup_site_eq_sum_buildings	site cost/usage rollup	= Σ child building rollups
rollup_customer_eq_sum_sites	customer rollup	= Σ site rollups
provider_rollup	per-provider totals	= Σ that provider's account bills
interval_counts	15-min channels over the span	full-period count (96/day) minus the deliberate missing_interval_backfill gap, restored after backfill
peak_demand_matches	VED demand line kW	= max 15-min kW in the period for interval buildings
monthly_eui_by_building	rp_MonthlyEUI	usage ÷ gross floor area, non-null for the 11 buildings with an area in the register (Campus Grounds and Cedar Row Carport and Grounds have none)
aggregate_campus_kwh	campus electric aggregate	= Σ member building interval kWh
digital_on_hours	rp_DigitalSummary	chiller/occupancy/exhaust on-hours per month, non-zero
notes_images_present	rp_NoteHistory / rp_ImageManagement	seeded notes + bill images returned
late_bill_detected	rp_LateBills	Sierra Gas Library late fixture present
rate_change_boundary	VED bills ≥ 2025-01	use FY2025 rates (higher than 2024)
meter_replacement_continuity	Library electric across 2024-08-14/15	no double-count; continuous reads
estimated_then_corrected	Sierra Gas Student Center	estimate superseded by actual; no double-count
duplicate_import_no_double	RCU Cedar Row A re-import	no service line double-counted

How to use

- Treat the **relationship** assertions (reconciliations, rollup equalities) as generator-independent invariants — they must hold for any seed.
- Treat the **fixed-count** assertions (calendar 24, security 27/19/18) as exact.
- As phases 2–6 land, extend this file (rate-engine determinant results, AP file contents, tenant allocation reconciliation, GHG totals, etc.).